

Vector Search Systems Fail on Complex Queries

Query("1947 Science Linguistics books") → Perform Vector Similarity Search → Top-k matches



Similarity Score = 0.675
(False Negative)

"... Meaning and Necessity: A Study in Semantics and Modal Logic (1947; enlarged edition 1956) is a book about semantics and modal logic by ..."



Similarity Score = 0.675
(False Negative)

"... Educating Eve: The 'Language Instinct' Debate (1999) is a book by Geoffrey Sampson critiquing evolutionary linguistics and language instinct theories ..."

Semantic Similarity ≠ Query Relevance

- Embeddings excel at capturing broad semantic relatedness, not precise intent.
- Vector search struggles with implicit predicates, constraints, and set operations.

Hybrid Search Requires Manual Labor

- Users must manually curate metadata fields and secondary indexes (e.g., BM25).
- Query-to-plan translation is often handwritten or LLM-driven and fragile.
- **Missed opportunity:** We lack automation and cost-based optimization.

Research Questions

Bootstrapping Hybrid Search

- How can we reduce the "cold-start" problem and quickly converge on a feasible index and query-planning design?

Caching Query Plans

- How can we avoid invoking an LLM-based rewriter for every new query?

Multi-Objective Learning

- How to propagate quality signals from downstream to improve index design and validate assumptions?
- How to apply learning to the LLM rewriter?

System-Level Optimizations

- How to build the optimal data engine to serve hybrid queries?

A New Architecture for Hybrid Search

Instance-Optimized Index Management

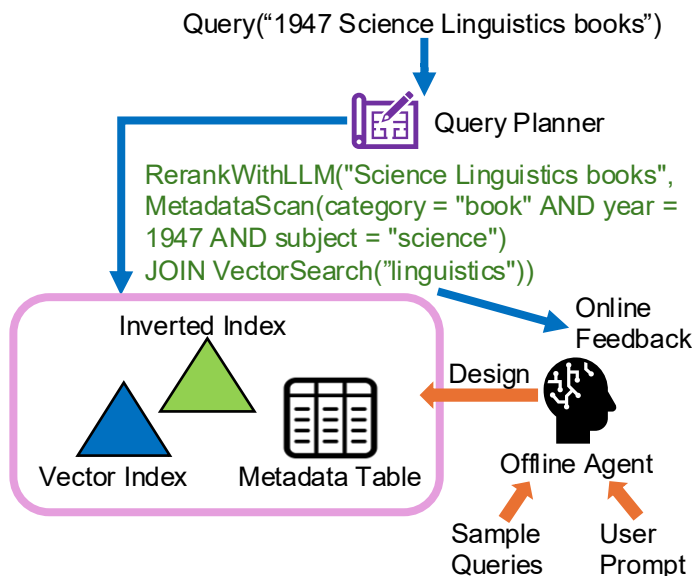
- Given query traces, automatically extract metadata columns and build sparse/dense retrieval indexes.
- Users provide space constraints and simple natural-language hints.

Automatic Query Planning

- At query time, natural-language search terms are translated into a hybrid logical plan.
- Plans combine metadata filters, vector search, and keyword search in a single execution strategy.

Learned Query Optimization

- Cost-based optimization maps the logical plan to a physical plan using learned statistics.
- Multi-objective optimization over quality (recall/precision) and performance, extending traditional query optimization.



Project Roadmap

Phase 1: LLM Rewriter + Database Designer on Existing Vector DB

- Chat with us if you have workloads or ideas!

Phase 2: Integration with Semantic Operators

- Can this be the indexing layer to complement PZ's scans?

Phase 3: Deep Integration with Vector DB Internals

- Joint indexes and optimized traversals
- Improved statistics tracking
- Learned data layouts
- Leveraging correlations
- Many more opportunities!